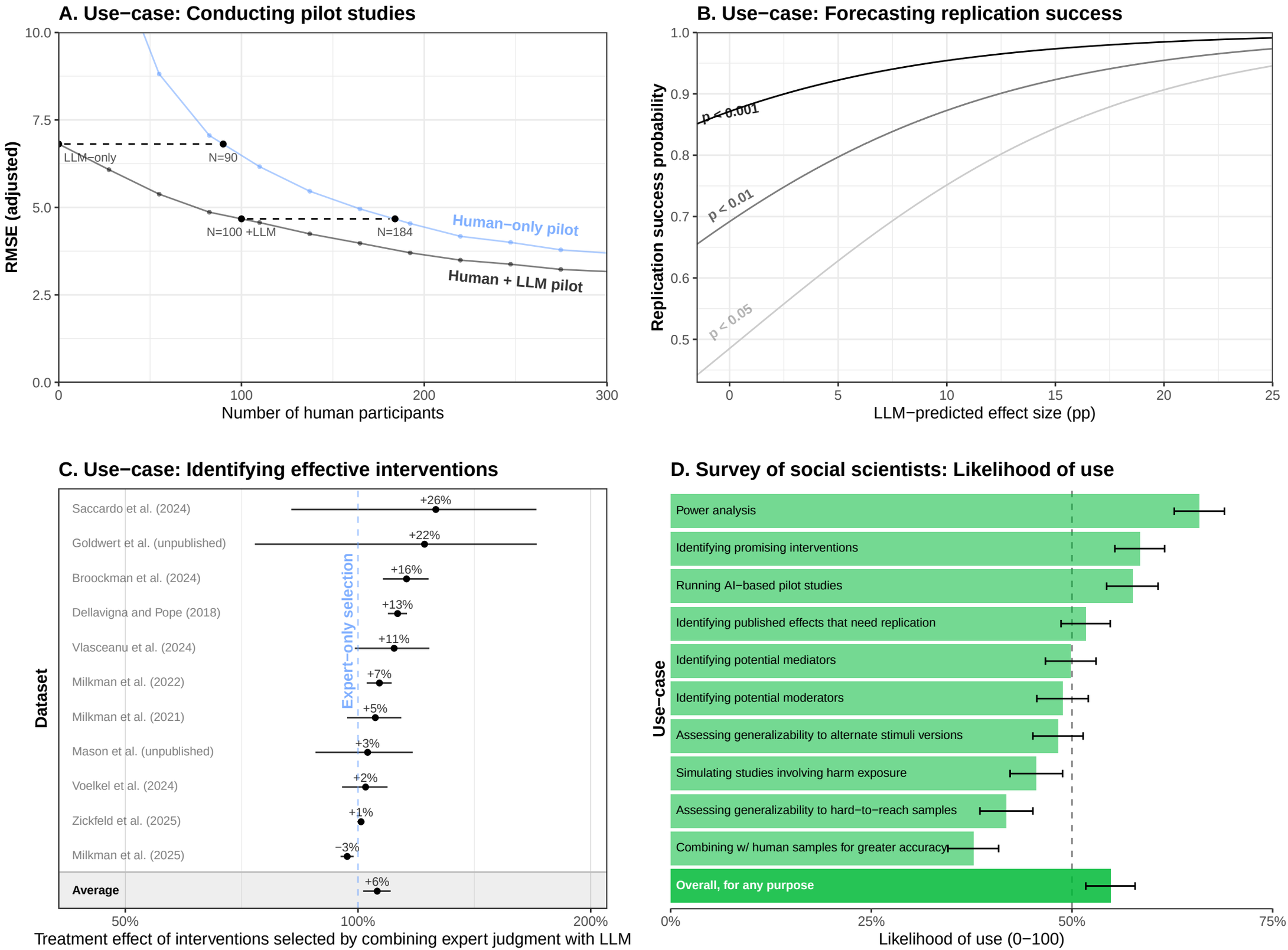
A. Use−case: Conducting pilot studies
RMSE (adjusted)
10.0
7.5
5.0
2.5
0.0
0
100
200
300
Number of human participants
LLM−only
N=90
N=100 +LLM
N=184
Human−only pilot
Human + LLM pilot
B. Use−case: Forecasting replication success
Replication success probability
1.0
0.9
0.8
0.7
0.6
0.5
p < 0.001
p < 0.01
p < 0.05
0
5
10
15
20
25
LLM−predicted effect size (pp)
C. Use−case: Identifying effective interventions
Dataset
Saccardo et al. (2024)
+26%
Goldwert et al. (unpublished)
+22%
Broockman et al. (2024)
+16%
Dellavigna and Pope (2018)
+13%
Vlasceanu et al. (2024)
+11%
Milkman et al. (2022)
+7%
Milkman et al. (2021)
+5%
Mason et al. (unpublished)
+3%
Voelkel et al. (2024)
+2%
Zickfeld et al. (2025)
+1%
Milkman et al. (2025)
−3%
Average
+6%
Expert−only selection
50%
100%
200%
Treatment effect of interventions selected by combining expert judgment with LLM
D. Survey of social scientists: Likelihood of use
Use−case
Power analysis
Identifying promising interventions
Running AI−based pilot studies
Identifying published effects that need replication
Identifying potential mediators
Identifying potential moderators
Assessing generalizability to alternate stimuli versions
Simulating studies involving harm exposure
Assessing generalizability to hard−to−reach samples
Combining w/ human samples for greater accuracy
Overall, for any purpose
0%
25%
50%
75%
Likelihood of use (0−100)